

KERA
Kerala Climate Resilient Agri-Value Chain Modernization
IRRI
CGIAR

KERA-AWD Connect

Insights from research and field actions from Kerala's low-emission rice systems

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Getting Ready for Mundakan 2026-27

RACING AGAINST WATER: FARMING IN THE KOLE

In the low-lying Kole lands of Thrissur, every single rice season starts with a race against water. Because these fields sit below sea level, managing moisture is a constant balancing act. It is not just about too much water; farmers face the opposite extreme just as often. When canal water releases are irregular or delayed, drought-like stress sets in during critical crop growth stages.

Beyond water availability, the challenges compound across the season. Bunds breach, canals silt up, pumping capacity is limited, and rising salinity poses a constant threat: last year, salt water seeped into the Vengidangu fields after water levels in the Enamakkal Phase Canal dropped too low. Even when farmers make it to harvest, sudden rains can flatten crops, labor is scarce, and heavy harvesters sink into the soft, muddy soil, leaving fields waterlogged for months afterward.

Despite all these challenges, the determination of these farming communities always shines through. Even the name Kole means a "bumper yield," reminding us how sweet the victory is when they successfully manage the water and harvest a rich crop.

In the Kole, water is at once the greatest challenge and the greatest opportunity. This duality forms the core focus of the KERA-AWD initiative: optimizing water efficiency, mitigating greenhouse gas emissions, and implementing climate-responsive rice cultivation practices tailored directly to the realities of the farmer's field.

STRENGTHENING THE EVIDENCE BASE

1. On-Station Evaluation of Low-Emission Rice Systems under Different Crop ManagementPractices

Recent on-station rice experiments are advancing sustainable agriculture by evaluating diverse crop establishment methods-including broadcasted, drum-seeded, and transplanted systems-alongside innovative nitrogen management practices like urea deep placement and top dressing. To capture a complete picture of crop performance and environmental impact, researchers are deploying cutting-edge tools, including Green Seeker and SPAD meters, sensor installations, PANI pipes, and continuous greenhouse gas (GHG) monitoring via specialized chambers and a Trace Gas Analyzer (TGA). This comprehensive approach is further supported by regular soil and destructive plant sampling, biometric observations, and crop monitoring across all growth stages to link nutrient efficiency and yield performance directly with GHG emissions.



fig 01. Taking readings with a handheld GreenSeeker at RARS, Pattambi



fig 02. Leaf greenness being checked with a SPAD meter



fig 03. Urea deep placement in the field at ARS, Mannuthy



fig 04. Soil sample collection before top dressing

2. Advances in Greenhouse Gas Monitoring and Analysis

Station trials at ARS Mannuthy and ARS Pattambi have achieved a significant milestone in GHG sample analysis, with 9,626 samples processed so far utilizing Gas Chromatograph (GC) facilities. This analysis spans three primary experiments across different irrigation regimes: Experiment 1 focusing on soil amendments, Experiment 2 examining different rice cultivars, and Experiment 3 exploring soil microbiomes. These studies provide crucial insights into how distinct soil treatments, varietal differences, and microbiological conditions influence GHG emissions. With roughly 1,300 samples left to process, the work is progressing systematically toward a September 2026 completion target, ensuring thorough data processing, treatment comparison, and consistent documentation.



fig 05. GHG sample analysis using the Shimadzu Nexus gas chromatograph

3. Advancing Climate-Smart Research: Launch of the KERA Ph.D. Network

Following high-level consultations and deliberations with KERA, University leaderships and global academic partners like IRRI, Cornell University (USA), Wagenigen University (Netherlands) and Imperial College(London), the KERA Ph.D. and early-career research network has been successfully established.

The program has selected its first cohort of 9 Ph.D. scholars from KAU across Agronomy (3), Agricultural Entomology (2), Agricultural Extension (3), and has now been unified to support 25 total scholars across two cohorts under a single category, featuring co-supervision and global research exchange opportunities aligned with project themes such as GHG mitigation and landscape hydrology.

THE FIRST COHORT OF KERA PH.D. SCHOLARS



Hilal Mohamed
Agronomy

Agrotechniques for mitigating greenhouse gas emission in transplanted rice



Anand R Das
Agronomy

Morphology and management of Isachne miliacea in wet-seeded rice (Oryza sativa L.)



Bismitha S N
Agronomy

Water hyacinth derived nanocellulose composite coated urea for improved nitrogen use efficiency in rice (Oryza sativa L.)



Aminu Sulhath T A
Agricultural Entomology

Influence of alternate wetting and drying (AWD) on pest dynamics in rice



Sala Srivarsha
Agricultural Entomology

Pesticide mediated changes in methane-cycling microbial communities in soil and insect gut under different water management regimes in rice



Anuranj P R
Agricultural Extension

Community nudges and ecological emotions for low emission rice systems in Kerala



Keerthana Vijayan
Agricultural Extension

Agroecology performance and climate resilience evaluation of flagship government programmes in Kerala.



Razia Fathima
Agricultural Extension

Climate induced migration in Kuttanad: a multidimensional analysis



Sreelakshmi K S
Agrometeorology

Greenhouse gas flux dynamics through process-based modelling and climate risk assessment in rice ecosystems of Kerala

CONNECTING RESEARCH TO GRASSROOTS

1. Assessing Emission Dynamics in Flooded Kole Fields



fig 06. Water scale fixed in a Kole field



fig 07. Plot selection for treatment plots in the Kole lands

During the last quarter (July – September 2026), the Kole fields remained flooded due to seawater intrusion, providing an important field setting for GHG monitoring. Weekly monitoring assessed methane, nitrous oxide, and carbon dioxide emissions using a TGA, smart chamber, and customized floating device, with three gas-sampling replications established alongside water-level meters in each field to record weekly fluctuations. Concurrently, field surveys were conducted to identify treatment plots for the upcoming season, marking an exciting step forward as KERA-AWD successfully extends its footprint to two new padasekharams, namely Elamutha and Ponnamutha in Vengidangu, for the upcoming Mundakan season. This expansion into new areas broadens the project's impact, capturing unique hydrological settings and emission dynamics. Meanwhile, Kole padasekharams are actively preparing for the next cropping season, with nursery beds already sown in some areas and seed arrangements and preparatory activities underway in others.

2. From Sowing to Harvest: Monitoring Palakkad Fields

Field-level monitoring and crop observations are actively progressing across multiple experimental and farmer fields. Soil moisture measurements and GHG observations utilizing the TGA are being conducted in both treatment and control plots across several locations. Furthermore, base and sensor installations established at designated sites are being monitored, complemented by periodic biometric observations to evaluate crop growth and development. Concurrently, harvesting has commenced in some padasekharams, with detailed harvest observations being systematically recorded to assess overall crop performance and key yield parameters under varying field conditions. During the Virippu season, these monitoring activities successfully covered six key padasekharams: Uppumpadam, Uppumpadam Central, Vakkad, Thuppalam, Kannampully 1, and Kannampully 2.



fig 08. One-cent harvest at Kannampully padasekharam

3. Empowering Farmers for Mundakan 2026–27: Bridging Knowledge and Practice

As part of preparations for the Mundakan season 2026–27, farmer sensitisation cum training programmes were initiated in the last week of July to enhance awareness of innovative agricultural technologies and improve access to project benefits and services. A total of six programmes covering nine padasekharams in Palakkad and five programmes covering six padasekharams in Thrissur were organised. Led by experienced faculty members and subject experts from the KAU, the sessions featured practical guidance and hands-on demonstrations on improved farming practices.

on improved farming practices. Farmers actively participated by engaging with resource persons, sharing field experiences, and seeking clarifications, creating a valuable platform for knowledge exchange that supports informed decision-making and technology adoption for the upcoming season.



fig 09. Sensitisation programme at Onpathmuri Kole padasekharam



fig 10. Sensitisation programme at Kattussery, Palakkad

4. Connecting Farmers to Benefits: KATHIR Registration and Geotagging Drive



fig 11. KATHIR registration of farmers

Alongside the sensitisation programmes, KATHIR registration and geotagging of farmers' plots were initiated in the last week of July across participating padasekharams as part of project activities. This drive aims to establish a systematic digital record of farmers and their cultivated holdings to facilitate seamless access to project benefits and services. Incorporating plot geotagging for precise documentation, the initiative has garnered enthusiastic participation from the farming community, with nearly 700 farmers successfully completing KATHIR registration so far as the process continues across the project area.

5. Cultivating Change, Nurturing Progress: KERA-AWD Expands to New Fields

The KERA-AWD project has extended its field-level interventions to the Elamutha and Ponnamutha padasekharams in Vengidangu Grama Panchayat, Thrissur, further advancing efforts to promote low-emission and climate-responsive rice cultivation. This expansion forms a key part of the scaling strategy for the Mundakan 2026–27 season, incorporating Elamutha (~300 acres) as the experimental site and Ponnamutha (~275 acres) as the control area to establish an experimental-control pair. Additionally, the scaling strategy includes the State Seed Farm, Alathur in Palakkad (6 experimental acres and 7 control acres), providing an opportunity to evaluate low-emission practices across conventional, natural, and organic farming systems.

These new sites add a vital dimension to ongoing efforts toward the Carbon Neutral Manalur vision. With this addition, KERA-AWD now operates in the Elamutha and Ponnamutha padasekharams alongside the existing six Kole padasekharams, generating robust evidence on performance under both farmer-managed and institutional conditions.



fig 12. Hon. MLA of Manalur, Prof. C. Raveendranath, inaugurating the sensitisation programme at Vengidangu



fig 13. Sensitisation programme for Elamutha and Ponnamutha farmers at Vengidangu

To support this expansion, dedicated farmer sensitisation programmes were organized to properly train and inform participants. Through comprehensive technical sessions, farmers were introduced to the principles and practical application of AWD, emphasizing better water management, low-emission farming techniques, and improved farming techniques, and improved cultivation practices. These sessions ensured that growers in the newly integrated padasekharams were well-equipped to connect the technology to their own fields, while simultaneous KATHIR registrations helped streamline digital data management for the upcoming season.

GEOSPATIAL PROGRESS: MAPPING THE LANDSCAPE FOR CLIMATE-RESILIENT RICE SYSTEMS

Geospatial activities under KERA-AWD continued to strengthen the spatial understanding of agricultural landscapes in Palakkad and Thrissur. During this quarter, significant progress was made in soil sampling, Land Use/Land Cover (LULC) mapping, drone-based mapping and ground-truth data collection, supporting the development of a comprehensive spatial database for the agricultural landscapes.

1. Soil Sampling: Building the Foundation for Digital Soil Mapping

Soil sampling was completed across Palakkad and Thrissur districts to support digital soil mapping. A total of 243 soil samples were collected, comprising 148 from Palakkad and 95 from Thrissur. Samples were gathered across four specific soil depths: 0–5 cm, 5–15 cm, 15–30 cm, and 35–45 cm. Laboratory analysis of these collected samples is currently underway.

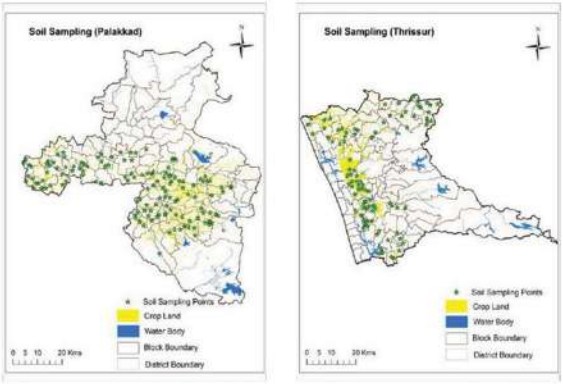


fig 14. Locations of soil samples collected in Palakkad and Thrissur

2. Land Use/Land Cover Mapping: Understanding the Agricultural Landscape

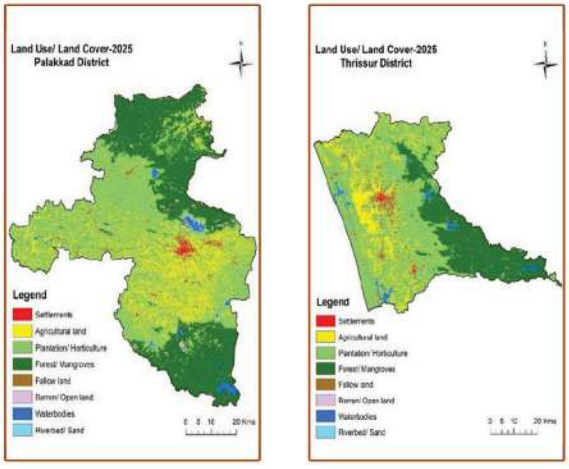


fig 15. Land Use/Land Cover maps of Palakkad and Thrissur

LULC mapping for the Palakkad and Thrissur districts was executed utilizing satellite remote sensing data to precisely delineate and quantify major land-use and land-cover classes across the study area. The entire region was classified into eight primary LULC categories, and the generated maps illustrate the spatial distribution and relative extent of these distinct classes across both districts.

3. Drone Mapping: Capturing the Field-Level Landscape

Drone-based mapping was undertaken to generate high-resolution spatial information and support detailed area delineation at selected agricultural locations. Mapping was carried out at the Elamutha Kole lands in Vengidangu, Thrissur, and at the State Seed Farm, Alathur, Palakkad. The high-resolution imagery from these surveys gives detailed field-level information for spatial assessment and documentation of agricultural areas.



fig 16. Area delineation and drone mapping at Elamutha Kole lands, Vengidangu, Thrissur

4. Ground Truthing: Strengthening Mapping Accuracy



fig 17. Ground-truthing activities at Pattambi and Thrithala blocks, Palakkad

Ground-truth data collection was systematically executed across the Palakkad and Thrissur districts to validate and enhance the accuracy of crop-type and LULC classification. Field-level observations were gathered from designated locations to support satellite-based image interpretation, with specific ground-truthing activities carried out across the Pattambi and Thrithala blocks of Palakkad.

FOSTERING FOUNDATIONS, ADVANCING GROWTH

1. Equipping the Project Team for Better Farmer Services

A dedicated KATHIR training programme was organised for project staff to enhance digital competencies and ensure efficient farmer registration and service delivery. Specialists from the Department of Agriculture instructed the team on the functionalities and protocols of the KATHIR platform, covering the registration of new farmers, the addition of agricultural plots, plot geotagging, and the updating of crop particulars within the system. The session integrated technical instruction with practical, hands-on exercises, enabling participants to directly execute various KATHIR activities.



fig 18. Hands-on session during the KATHIR training for project staff

2. Strengthening Technical Expertise: Training on Trace Gas Analyzer

A specialized training programme on the TGA was conducted to enhance the technical proficiency and operational capabilities of the project team in trace gas measurement and data acquisition. Led by experts from LI-COR Environmental Solutions, including experienced scientists and senior technicians, the training combined comprehensive technical instruction with live demonstrations covering the fundamental working principles and operational aspects of both the TGA and the Smart Chamber. Participants were guided through equipment configuration, system setup, and standard operating procedures for field-level measurements, gaining valuable hands-on experience in hardware handling. The programme placed special emphasis on data retrieval, downloading, and management protocols to ensure efficient organization of field datasets, while interactive discussion segments provided an opportunity for attendees to resolve technical queries directly with the specialists.



fig 19. Training programme on the functioning of the Trace Gas Analyzer

3. Decoding Climate Risks: Training Workshop on Compound Extreme Events

A specialized training programme on "Impacts of Compound Extreme Events on Agriculture: Concepts, Assessment and Applications" was hosted on 3rd August 2026 at Kelappaji College of Agricultural Engineering and Technology (KCAET), Tavanur, by Kerala Agricultural University (KAU) and the International Rice Research Institute (IRRI). The workshop brought together 49 participants including faculties, students, and scientists to focus on assessing how simultaneous climate hazards impact agricultural productivity and water resources. Through technical sessions led by experts Dr. Shailza Sharma (Assistant Professor, IIT (ISM) Dhanbad) and Dr. Proloy Deb (Scientist, IRRI), attendees explored foundational concepts, agricultural modeling frameworks, and analytical methods.

The programme concluded with an afternoon practical session led by expert Dr. Shailza Sharma, where researchers analyzed regional meteorological data from Kerala to identify and quantify compound extremes using R programming.



fig 20. Training workshop on compound extreme events at KCAET, Tavanur

4. Mapping Canal Infrastructure: High-Precision DGPS Survey Along the Malampuzha Left Bank Canal

A high-precision Differential Global Positioning System (DGPS) survey was initiated along the Left Bank Canal of Malampuzha Dam by the Centre for Water Resources Development and Management (CWRDM). Utilizing advanced satellite signals and reference station corrections to determine accurate ground coordinates and elevations, the survey gather essential data on canal cross-sections and elevation variations.

The resulting insights will help evaluate the canal's geometric characteristics, providing critical inputs for developing a Decision Support System (DSS) to enhance canal operation, irrigation water management, and distribution planning.

5. Strengthening Global Collaboration: Highlights from the KAU Delegation's Visit to IRRI HQ

The visit of the KAU delegation, headed by the Vice Chancellor and comprising the Director of Research, Associate Director of Research, Director of Students' Welfare and other faculty members, to the IRRI headquarters in the Philippines from 16–21 September under the KERA project featured several important highlights. These included exploring the International Rice Genebank's sub-zero cold storage facilities for seed conservation, touring the D.L. Umali Laboratory, and visiting the Kenzo Hemmi Laboratory to study low-glycemic index rice breeding aimed at mitigating Type-2 diabetes. The team also examined advanced GHG emission setups at the N.C. Brady Lab, reviewing multiplexer-enabled gas flux equipment for tracking nitrous oxide emissions, alongside field plots testing bio-stimulants, iron powders, and direct-seeded rice (DSR) cultivation methods. Future plans and undertakings discussed focused on expanding collaborative research support with IRRI scientists and advancing low-emission rice production strategies, building upon the ongoing technical assistance provided by the institute. Notably, the delegation held interactive discussions with IRRI Director General Dr. Yvonne Pinto and other senior officials, where mutual research insights were exchanged and frameworks for further support from IRRI were discussed. Additionally, the team connected with the National Irrigation Administration (NIA) Region IV-A and the Santa Maria Irrigation Association, aiming to understand grassroots water management through AWD practices, farmer interactions, and local canal maintenance approaches



fig 21. Discussions during the KAU delegation's visit to IRRI HQ



fig 22. Laboratory visit at IRRI HQ

Voices from the Field



Prof. C. Ravindranath

MLA, Manalur

Alternate Wetting and Drying is a vital climate-smart intervention for Kerala's paddy cultivation, addressing water scarcity and greenhouse gas emissions. The active collaboration of farmers, Padasekharam committees, and government departments is essential to bridge scientific knowledge with field realities. Manalur's paddy fields hold immense potential to model sustainable, profitable, and water-efficient rice farming. I congratulate the KERA–AWD team for uniting these vital voices to secure a resilient agricultural future.



Ms. Nima

Agriculture Officer, Krishi Bhavan, Malampuzha

Climate change is reshaping the risk profile of paddy production, making efficient water management increasingly central to climate resilience. AWD strengthens this resilience by enabling farmers to schedule irrigation based on actual field conditions rather than following fixed irrigation routines. Supported by sustained extension services and reliable water-level monitoring, AWD empowers farming communities to adapt to changing rainfall patterns, use water more efficiently, and sustain paddy productivity under increasing climatic uncertainty



Mr. Shahul Hameed

Farmer, Uppumpadam Padasekharam

My farming journey, which began with memories of traditional agriculture, transformed into an entirely new experience through the KERA project and IRRI. Being introduced to modern paddy cultivation methods and scientific knowledge like AWD was a great realization. These kind of projects give me the opportunity to understand new technologies with a commitment to farmers and test them in my fields.

Meet the Team Behind KERA



Dr. Panneerselvam Peramaiyan

Scientist II – Cropping Systems Agronomy
IRRI South Asia Regional Centre (ISARC), Varanasi

Dr. Panneerselvam Peramaiyan is a Scientist II – Cropping Systems Agronomy at the International Rice Research Institute (IRRI) South Asia Regional Centre (ISARC) in Varanasi, India. He holds a Ph.D. from Aarhus University and leads research on direct-seeded rice (DSR), precision nutrient management, climate-resilient systems, and soil health. Formerly a Systems Agronomist at CIMMYT, his current work focuses on scaling sustainable agricultural innovations across India through partnerships with research, extension, and private sectors.



Dr. Shyla Joseph

Professor, Department of Soil and Water Conservation Engineering
Kerala Agricultural University

Dr. Shyla Joseph is a Professor in Soil and Water Conservation Engineering at Kerala Agricultural University, with a Ph.D. in Soil and Water Conservation and 27 years of experience in teaching and research. Her areas of specialization include agricultural water management, precision agriculture, micro-irrigation, and automated micro-irrigation systems. Her research focuses on developing innovative, technology-driven solutions for efficient irrigation water management and sustainable agricultural production. Her major research projects include an IoT-based automated drip irrigation system and an IoT-based discharge measurement system for canals, demonstrating the application of smart technologies in irrigation automation and water resource monitoring. With more than 26 research paper publications, she has contributed to the advancement of agricultural water management, precision irrigation, and efficient water-use technologies. Her professional interests centre on integrating automation, digital technologies, and modern irrigation practices to improve water-use efficiency, optimize irrigation management, and promote sustainable and climate-resilient agricultural systems.



Ms. Priya Vallat

Project Manager (Central Region), KERA

Ms. Priya Vallat is currently serving as Project Manager (Central Region) for the KERA project, bringing over 16 years of dedicated field experience across agricultural credit, public extension in Kerala. Beginning her career in 2010 as an Agricultural Officer with KSCARD Bank across Alathur, Chittur, and Palakkad taluks, she transitioned in 2020 to the Department of Agriculture and Farmers Welfare, serving in Ananganadi and Kuzhalmannam panchayaths of Palakkad District, alongside multiple tenures holding charge as Assistant Director of Agriculture for Kuzhalmannam Block. Managing extension in an intensive command area under the Malampuzha Left Bank Canal—where roughly 14,000 farmers cultivate paddy across the Viruppu and Mundakan seasons in Kuzhalmannam block of Palakkad District—gave her firsthand insight into water distribution challenges and farmer constraints. Leveraging this deep institutional knowledge and grassroots rapport, Priya played an instrumental role in advancing KERA Alternate Wetting and Drying (AWD) initiative by identifying suitable padasekharams and mobilizing farming communities across various stages of field-level implementation



Dr. Naveena K.

Scientist, Land and Water Management Research Group
Centre for Water Resources Development and Management (CWRDM), Kozhikode

Dr. Naveena K. is a Scientist in the Land and Water Management Research Group at the Centre for Water Resources Development and Management (CWRDM), Kozhikode, Kerala. He holds a Ph.D. in Agricultural Statistics from Banaras Hindu University and specializes in machine learning, time-series forecasting, and advanced data analytics for agricultural water management. His research applies statistical and computational approaches to address challenges in water security, climate change, agricultural productivity, and sustainable water resource management. As Principal Investigator, he has led research projects on water security assessment, hydrological model uncertainty quantification, climate change vulnerability in agriculture, and water-stress hotspot identification in Kerala. Dr. Naveena has authored books on rainfall trend analysis and R programming for hydrology and published over 20 Scopus-indexed journal articles. His accolades include the Young Scientist Award (2015), Excellence in Research Award (2017), and ICAR Senior Research Fellowship.



Mr. Akshay N.V.

Project Associate, KERA

Akshay N.V. holds a Master's degree in Agriculture with specialization in Agronomy from Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences, Shivamogga. He is currently working as a Project Associate under the KERA project where he is involved in low-emission rice production systems. He has participated in the adaptation of agronomic field trials in the Kole region of Thrissur and at the Agricultural Research Station (ARS), Mannuthy. His responsibilities include greenhouse gas (GHG) monitoring, field data sampling and collection, agronomic observations and field-level implementation of research trials. Prior to his current role, he worked as a Data Enumerator for the KERA project in Kannur and as a Field Technical Assistant at the Kissan Call Centre. His professional interests include agronomy, nutrient management, weed management, sustainable rice production and field-based agricultural research.

Mr. Bijith V.V.

Skilled Assistant, KERA–AWD Project



Mr. Bijith V.V. holds a B.Sc. in Zoology and a Diploma in Agricultural Science and serves as a Skilled Assistant in the KERA-AWD project. With an academic background in biological sciences and agriculture, he contributes to field research and monitoring activities in the Kole wetlands of Kerala. His responsibilities include greenhouse gas emission monitoring using a Trace Gas Analyzer (TGA) and water management sensor readings. Through his technical support and field-level observations, he contributes to agricultural research and the implementation of Alternate Wetting and Drying (AWD) practices, supporting sustainable and climate-resilient rice cultivation.



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